

DAY - **22**

SEAT NUMBER

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2026

III

10

1100

V - 738

(E)

**MECHANICAL MAINTENANCE
PAPER - II (A-2)**

Time : 2½ Hours**2 Pages****Max. Marks : 40****Instructions :** (1) All questions are compulsory.

(2) Figures to the right indicates full marks.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Use of drawing instruments, mathematical tables etc. is permitted.

(5) Assume suitable data, if necessary.

1. Answer the following : (Any Four)

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Fill in the blanks

(a) In 2-stroke engine working cycle is completed in _____ of crankshaft revolution.

(i) 360°

(ii) 720°

(b) Air vessel is used in _____ pump.

(i) Centrifugal

(ii) Reciprocating

Match the pairs :

(c) Bevel gears : (i) Connect the parallel shafts

(d) Helical gears (ii) Reduce the speed

(iii) Connect the perpendicular shafts

State 'true' or 'false' and correct it if false :

(e) Air compressor sucks air from atmosphere.

(f) In petrol engine ignition takes place at constant pressure.

2. Answer the following in about 8 to 10 lines each : **(Any Two)** 6
 - (a) Explain Carburettor with a neat sketch.
 - (b) Draw any six sketches of Pipe Fittings.
 - (c) Describe the working of Air Compressor with a neat sketch.
 3. Answer the following in about 8 to 10 lines each : **(Any Two)** 6
 - (a) Explain Muff Coupling with a neat sketch.
 - (b) What are the purposes of Lubricant ?
 - (c) Differentiate between Petrol Engine and Diesel Engine.
 4. Answer the following in about 5 to 6 lines each : **(Any Three)** 6
 - (a) Write a short note on Multistage Pump.
 - (b) How care of I.C. engine should be taken ?
 - (c) Draw a neat labelled sketch of Pipe Vice.
 - (d) Write short note on Belt Conveyor.
 - (e) Differentiate between Open Belt Drive and Cross Belt Drive with sketches.
 5. Answer the following in about 5 to 6 lines each : **(Any Three)** 6
 - (a) Classify the Pumps.
 - (b) What is meant by Clutch ?
 - (c) Draw a neat sketch of Screw Jack.
 - (d) Why foundation is necessary for machines ?
 - (e) Write a short note on Cylinder.
 6. Attempt **any two** : 12
 - (a) Explain the working of 4-stroke diesel engine with neat sketches.
 - (b) State the types of Gears. Explain any one with a neat sketch.
 - (c) What are the types of Impellers ? Explain any one with a neat sketch.
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